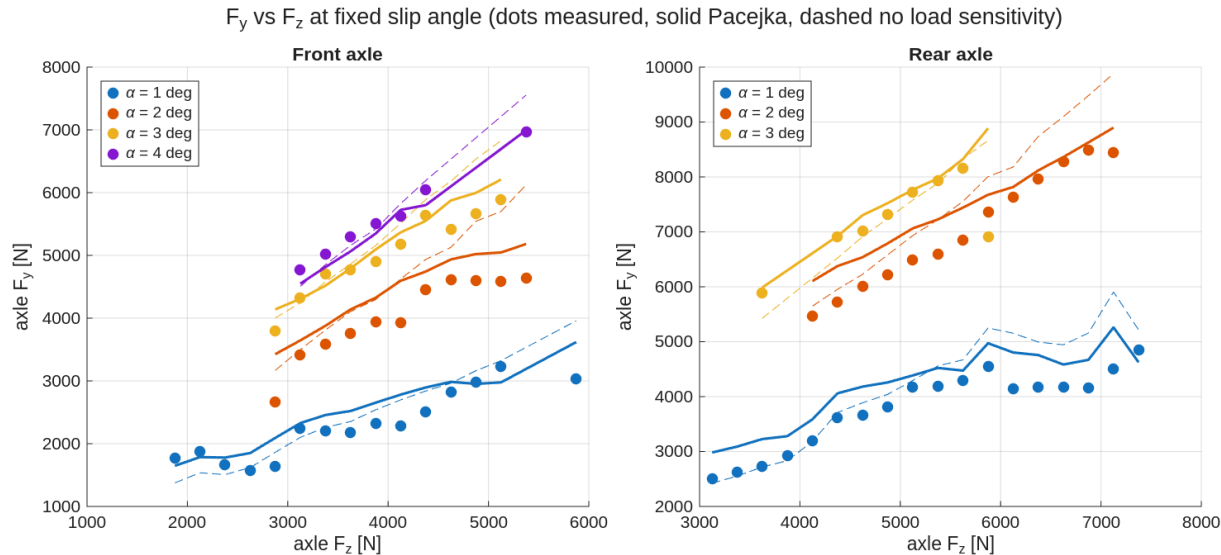


SLIDE: More load gives less grip per newton, and the tire model captures it

Put this plot on the slide (figure "Fy vs Fz at fixed slip angle" from tire_fz_plots.m) with the three bullets on the right.



Bullets for the slide

- **Same slip angle, more load: F_y rises less than F_z.** Front axle at 2°: F_z 3.1 kN gives F_y 3.4 kN. F_z 5.4 kN gives only 4.6 kN. That is 72% more load for 36% more force.
- **A tire with no load sensitivity would give 6.1 kN** (dashed lines). The measured points follow the load-sensitive Pacejka model (solid lines).
- **It holds on a log the model never saw.** The model was fitted on the 2025 Laguna log. On the 2026 Laguna log the measured slope is closer to Pacejka than to the no-load-sensitivity model in all 7 slip/axle cases.

Small print for the slide footer

Axle F_y from Kalman filter ($m \cdot a_y$ + yaw balance), F_z from load model + strain-gage observer. Near pure cornering only ($|a_x| < 2$, $|a_y| > 2$ m/s², $v > 10$ m/s), 35k samples, 2 Laguna Seca logs. Dots are medians of $\pm 0.25^\circ$ slip $\times$ 250 N load bins.

How to read it, and the numbers behind it

The idea (your x-slip, Fz1 → y1, Fz2 → y2 test). Pick samples where the axle runs at the same slip angle, then compare F_y at low load against F_y at high load. Load at the same slip angle changes mostly through speed (downforce), and also through banking, crests and load transfer. With no load sensitivity F_y is proportional to F_z , so the extra force per extra newton of load (dF_y/dF_z) equals F_y/F_z . A load-sensitive tire has a lower slope. The measured slope is the number that decides between the two models.

Worked examples (both logs, bin medians)

Axle, slip	Fz low	Fy meas	Fz high	Fy meas	Load +	Fy + meas	Fy + Pacejka	Fy + no load sens.
Front, 2°	3.1 kN	3.41 kN	5.4 kN	4.64 kN	+72%	+1.23 kN (+36%)	+1.54 kN	+2.63 kN
Front, 3°	3.1 kN	4.32 kN	5.1 kN	5.88 kN	+64%	+1.57 kN (+36%)	+1.90 kN	+2.58 kN
Rear, 2°	4.1 kN	5.46 kN	6.9 kN	8.48 kN	+67%	+3.02 kN (+55%)	+2.52 kN	+3.83 kN

Slope dF_y/dF_z at fixed slip angle (N per N, linear fit within each slip bin)

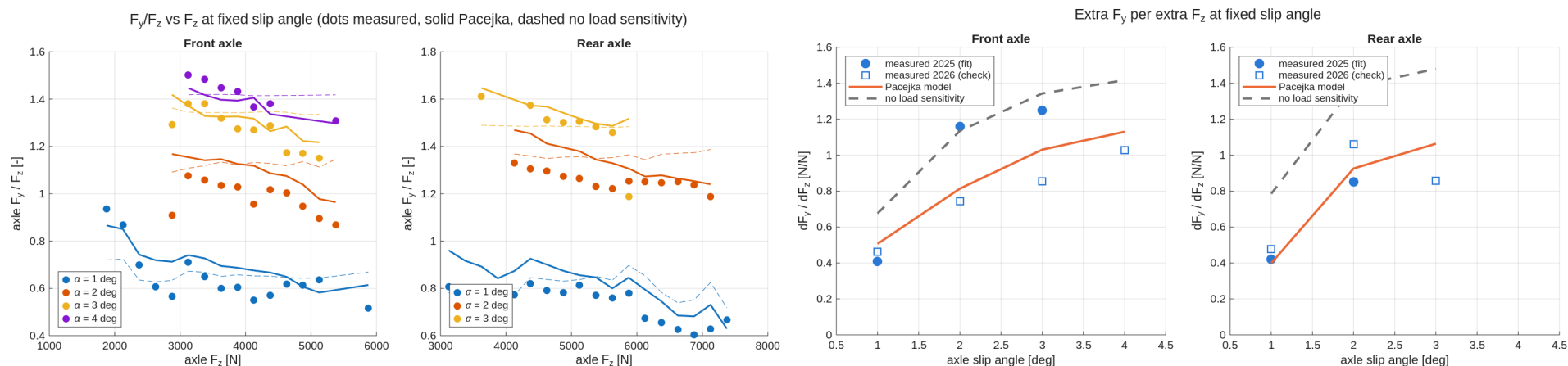
Axle	Slip	2025 meas (fit log)	2026 meas (new log)	Both logs meas	Pacejka	No load sens.
Front	1°	0.41	0.46	0.46	0.51	0.68
Front	2°	1.16	0.74	0.83	0.81	1.13
Front	3°	1.25	0.86	0.87	1.03	1.34
Front	4°	—	1.03	1.04	1.13	1.42
Rear	1°	0.42	0.48	0.45	0.40	0.79
Rear	2°	0.85	1.06	0.96	0.93	1.38
Rear	3°	—	0.86	0.84	1.06	1.48

Model slopes are from both logs together. "—" means fewer than 300 samples in the bin.

The fitted model (Pacejka 1987, 9 coefficients per curve)

Peak μ_y at tire Fz	1 kN	2 kN	3 kN	4 kN	5 kN	6 kN
Front 275/40R15	1.69	1.57	1.46	1.34	1.22	1.10
Rear 385/30R15	1.74	1.65	1.56	1.47	1.39	1.30

Backup slide and what to say if challenged



Left: the same data as F_y/F_z . With no load sensitivity (dashed) the lines are flat. The measured dots fall with load, as the Pacejka model (solid) does. **Right:** the table's slopes. Measured points mostly sit near the orange model line and well below the grey no-load-sensitivity line.

Honest limits (have these ready)

- **Neither force is a direct per-tire measurement.** Axle F_y comes from the Kalman filter ($m \cdot a_y$ + yaw balance, with I_z still a placeholder), and F_z from the load model plus strain-gage rate. The test shows axle-level load sensitivity. The per-tire split is inferred from it.
- **Overall error does not pick a winner.** Axle F_y RMS error is about 600–800 N for both models ($r = 0.89$ – 0.96). The slope at fixed slip is the part that separates them.
- **On its own, the 2025 fit log was mixed.** At front 2–3° its slope matched no load sensitivity, which is why the earlier report could not confirm it. The 2026 log has higher speeds (up to 61 m/s against 50 m/s) and so a wider load range, and it clearly shows the load-sensitive trend.
- **Load and speed are linked.** At a fixed slip angle, high load mostly means high speed (downforce). Tire temperature or aero balance changes with speed could produce part of the slope. To settle this, run constant-radius or ramp-steer tests at 3–4 speeds on one corner.
- **Level offset.** The model reads about 7–12% high on the front at 2°. The slope (the F_z effect) is right, but the grip level could be trimmed.
- **2026 wheel speeds were not calibrated** (not enough free-rolling samples). That affects F_x and slip ratio, not slip angle or F_y .